

SentinelPulse-LAR: Identity-Preserving Activation with Learnable Localized Corrections

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Abstract—Traditional neural network activations utilize fixed-shape nonlinearities that apply global transformations to input distributions. We propose SentinelPulse-LAR (Localized Activation Residual), an activation mechanism that treats nonlinearity as a learnable, localized perturbation of the identity mapping. By decomposing the activation into an identity path and a Gaussian-enveloped residual, we allow the network to learn specific local corrections governed by a per-activation amplitude (α) and a stage-shared locality parameter (β). Our results on CIFAR-100 and Yahoo Answers demonstrate that learnability of the residual amplitude is critical for performance gains. Furthermore, we posit that this mechanism serves as a primitive for local credit assignment, potentially reducing reliance on global backpropagation.

Index Terms—Activation Functions, Neural Architecture, Residual Learning, Local Credit Assignment, SPv5.

I. INTRODUCTION

The standard paradigm of activation functions such as ReLU [1], GELU [2], and Mish [3] relies on static mathematical curves. While effective, these functions are “blind” to the specific requirements of different layers, forcing the weight matrices to compensate for fixed nonlinear behaviors.

We introduce **SentinelPulse-LAR (SpLR)**, which reframes the activation as a *micro-residual* operation. Instead of replacing the input signal x , the mechanism “pulses” a correction only where needed. This design is motivated by the principle of minimal intervention: preserving the identity mapping by default and learning signed, localized deviations only when they improve representational capacity.

II. MECHANISM IDENTITY AND CLAIMS

We define SentinelPulse-LAR as a mechanism-focused contribution based on the following claims:

- **Claim 1:** Identity preservation in the tails ($|x| \rightarrow \infty$) prevents global representation distortion.
- **Claim 2:** The nonlinearity is explicitly localized via a learnable parameter β .
- **Claim 3:** Learnability of the residual amplitude α is a necessary condition for gains.
- **Claim 4:** The mechanism is structurally distinct and non-redundant compared to layer-level residual blocks.

III. SENTINELPULSE-LAR MECHANISM

A. Mathematical Definition

SentinelPulse-LAR is defined as:

$$y = x + \alpha \cdot x \cdot \exp(-\beta x^2) \quad (1)$$

The function decomposes into an identity path and a residual correction $r(x) = \alpha x e^{-\beta x^2}$.

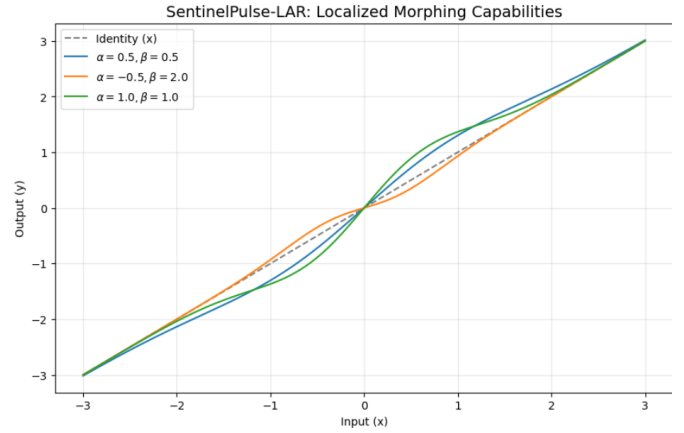


Fig. 1. Visual representation of SentinelPulse-LAR morphing capabilities. The function adapts its shape locally around the identity mapping based on learnable parameters α and β .

B. Parameterization and Golden Initialization

To ensure stability, we adopt a constrained parameterization:

$$\alpha = 2 \tanh(\alpha_{raw}), \quad \beta = 0.01 + \text{softplus}(\beta_{raw}) \quad (2)$$

We define the canonical “Golden Initialization” as $\alpha_{raw} = 0.182$ and $\beta_{raw} = -1.35$.

IV. EXPERIMENTAL RESULTS

A. Vision Domain: CIFAR-100

We evaluated SpLR against Mish using 3 seeds. SpLR consistently outperformed the baseline across both 25 and 50-epoch regimes.

B. Ablation of Learnability

To test Claim 3, we compared the canonical SpLR (A-0) against a shared amplitude variant (A-1) and a frozen amplitude variant (A-2).

TABLE I
MEAN TEST ACCURACY ON CIFAR-100 (3 SEEDS)

Activation	Epochs	Dropout	Val Acc (%)	Test Acc (%)
Mish	25	0.10	53.25 $\pm$ 0.21	56.17 $\pm$ 0.09
SpLR	25	0.10	54.50 $\pm$ 0.38	58.49 $\pm$ 0.21
Mish	50	0.10	56.26 $\pm$ 0.32	57.90 $\pm$ 0.30
SpLR	50	0.10	59.02 $\pm$ 0.22	61.75 $\pm$ 0.32

TABLE II
ABLATION STUDY RESULTS (CIFAR-100, 25 EPOCHS)

Variant	α Policy	Description	Acc (%)
A-0	Local	Canonical SpLR	35.26
A-1	Shared	Stage-shared amplitude	36.03
A-2	Frozen	No amplitude learning	29.81

V. DISCUSSION: RESULT ANALYSIS

A critical finding in our ablation studies is the 6.22% performance drop observed when α is frozen (A-2). This validates the core thesis that the network requires the ability to learn the magnitude of the localized correction. While the shared variant (A-1) showed a slight advantage in a low-epoch regime (25 epochs), we hypothesize that local per-activation learnability (A-0) becomes more critical as architectural depth and task complexity increase, allowing for finer-grained representational shaping.

VI. CONCLUSION AND FUTURE WORK

SentinelPulse-LAR introduces a robust, identity-preserving activation mechanism. Beyond empirical gains, SpLR provides the mathematical foundation for decentralized learning. Future work will focus on a dedicated optimizer that updates α based on local signal statistics, moving away from global backpropagation.

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